Eunkyu Han
Staff Scientist
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EDUCATION

Boston University
Ph.D. in Astronomy — July 2020
Thesis: The Mass-Radius Relationship of M Dwarf Stars from Kepler Eclipsing Binaries
Advisor: Prof. Philip S. Muirhead

The Pennsylvania State University
B.S. in Astronomy & Astrophysics with Honors — December 2012
B.S. in Physics
Minor in Mathematics
Senior Thesis: The Exoplanet Orbit Database
Advisor: Prof. Jason T. Wright

PROFESSIONAL POSITIONS

Staff Scientist II — October 2023 - Present
Space Telescope Science Institute
- Supporting the science development of the Wide Field Instrument (WFI) as a member of the Roman Telescope Branch within the Instruments Division
  - Contributing to the development of simulation tools
  - Serving as the product owner of the Roman WFI Exposure Time Calculator (ETC) in the multi-mission ETC team
  - Serving as a deputy block lead of the simulation tools block within the branch
  - Serving as a member of the simulation tools group at the Roman help desk

IGRINS postdoctoral fellow — August 2020 - August 2023
The University of Texas at Austin
- Measured magnetic field strengths of low-mass stars using high-resolution near-infrared IGRINS spectra
- Trained new IGRINS observers and supported IGRINS observations at Gemini-South

Graduate Research Assistant — January 2015 - July 2020
Boston University
- Determined a precise mass-radius relationship of M dwarfs from Kepler eclipsing binaries

- Conducted follow-up observations of M dwarf eclipsing binaries
  - High-resolution near-infrared spectroscopic observations with IGRINS & iSHELL
  - Near-infrared photometric observations with Mimir
- Searched for low-mass eclipsing binaries in TESS using machine learning algorithm

Graduate Teaching Fellow Fall 2014 - Spring 2015
Boston University
- Taught discussion sections and night labs for AS 105 (Alien Worlds) and AS 100 (Comic Controversies)

Research Assistant 2013 - 2014
Center for Exoplanets and Habitable Worlds, The Pennsylvania State University
- Investigated open cluster candidate Loden1 through radial velocity measurements from SALT spectra
- Spectroscopic studies of Ruprecht 147 white dwarfs and M dwarfs using SALT & GTC OSIRIS spectra

Undergraduate Researcher 2010 - 2012
The Pennsylvania State University
- Maintained and extended the Exoplanet Orbit Database and exoplanets.org

RESEARCH INTERESTS

- Determining detailed physical properties of low-mass stars
- Searching for and characterizing low-mass eclipsing binaries using space-based observations
- Near-infrared high-resolution spectroscopic observations of low-mass stars
- Studying characteristics of planet-hosting stars

PUBLICATIONS

First-Author Refereed Publications (total 6)

Han, E., López-Valdivia, R., Mace, G., Jaffe, D., “Magnetic field measurements of low-mass stars from high-resolution near-infrared IGRINS spectra”, *The Astronomical Journal*, 166, 4 (2023)

Han, E., Rappaport, A. S., Vanderburg, A., Tofflemire, M. B., Borkovits, T., Schwengeler, M. H., Zasche, P., Tokovinin, A., Krolikowski, M. D., Muirhead, S. P., Kristiansen, M. H., Terentev, I., Omohundro, M., Gagliano, R., Jacobs, T., LaCourse, D.
“A 2+1+1 quadruple star system containing the most eccentric, low-mass, short-period, eclipsing binary known” *Monthly Notices of thee Royal Astronomical Society*, 510, 2448 (2022)

Han, E., Muirhead, S. P., Swift, J., “Magnetic Inflation and Stellar Mass IV. Four Low-mass Kepler Eclipsing Binaries Consistent with Non-magnetic Stellar Evolutionary Models” *The Astronomical Journal*, 158, 111 (2019)

Han, E., Muirhead, S. P., Baranec, C., Law, N., Riddle, R., Atkinson, D., Mace, G., DeFelippis, D., “Magnetic Inflation and Stellar Mass I. Revised Parameters for the Component Stars of the Kepler Low-mass Eclipsing Binary T-Cyg1-12664” *The Astronomical Journal*, 154, 100 (2017)

Han, E., Curtis, J., Wright, J., “The Putative Old, Nearby Cluster Laden 1 Does Not Exist”, *The Astrophysical Journal*, 152, 7 (2016)

Han, E., Wang, S., Wright, J., Feng, Y., Zhao, M., Onsi, F., Brown, J., Hancock, C., “Exoplanet Orbit Database. II. Updates to Exoplanets.org” *Publications of the Astronomical Society of the Pacific*, 126, 827 (2014)

Co-Authored Refereed Publications (total 7)

López-valdivia, R., Mace, G., **Han, E.**, Sawczynec, E., Hernández, J., Prato, L., Johns-Krull, C., Oh, H., Lee, J., Kraus, A., Llma, J., Jaffe, D., “The IGRINS YSO Survey III: Stellar parameters of pre-main sequence stars in Ophiuchus and Upper Scorpius”, *The Astronomical Journal*, submitted

Hussaini, M., Mace, G., López-Valdivia R.,Honaker, E., and **Han, E.**, “The Impact of Rotation Velocity on Measuring Magnetic Fields of K and M Stars” *Research Notes of the American Astronomical Society,* 4, 241 (2020)

Honaker, E., Mace, G., **Han, E.,** “TESS Photometry of the Precataclysmic Variable Wolf 1130AB” *Research Notes of the American Astronomical Society,* 4, 1 (2020)

Healy, F. B., **Han, E.**, Muirhead, S. P., Skiff B., Polakis T., Rilinger A., and Swift, J. S., “Magnetic Inflation and Stellar Mass. III: Revised Parameters for the Component Stars of NSVS 07384765” *The Astronomical Journal,* 158, 89 (2019)

Winters, J., Irwin, J., Newton, E, Charbonneau, D., Latham, D., **Han, E.**, Muirhead, P., Berlind, P., Calkins, M., Esquerdo. G., “LHS 1610A: a Nearby Mid-M Dwarf with a Companion that is Likely a Brown Dwarf” *The Astronomical Journal*, 155, 125 (2018)

Croll, B. Dalba, P., Vanderburg, A, Eastman, J, Rappaport, S, DeVore, J, Bieryla, A, Muirhead, P., **Han, E.**, Latham, D., Beatty, T., Wittenmyer, R., Wright, J., Johnson, J., McCrady, N., “Multiwavelength Transit Observations of the Candidate Disintegrating Planetesimals Orbiting WD 1145+017” *The Astrophysical Journal*, 836, 82 (2017)

Wright, J., Onsi, F., Marcy, G., **Han, E.**, Feng, Y., Johnson, J., Howard, A., Valenti, J., Anderson, J., Piskunov, N., "The Exoplanet Orbit Database" *Publications of the Astronomical Society of the Pacific*, 123, 412 (2011)

OBSERVING EXPERIENCE & TELESCOPE TIME AWARDED

Principal Investigator

Gemini Observatory, Cerro Pachon, Chile 2018 - 2022
- 8.0-meter Gemini-South with IGRINS: 95 hours

NASA InfraRed Telescope Facility, Mauna Kea, HI 2017 - 2022
- 3.0-meter IRTF with iSHELL: 46 half-nights

McDonald Observatory, Fort Davis, TX 2021 - 2022
- 2.7-meter Harlan J. Smith Telescope with Robert G. Tull Coudé Spectrograph: 30 full-nights

Lowell Observatory, Flagstaff, AZ
- 4.3-meter LDT (formerly known as DCT) 2016 - 2020
  - With IGRINS: 30 half-nights
  - With DSSI/QWSSI: 10 half-nights
- Perkins 72-inch telescope with Mimir and PRISM: 58 full-nights 2015 - 2017

The W.M. Keck Observatory, Mauna Kea, HI 2013
- 10.0-meter Keck I with HIRES: 3 half-nights

Co-Investigator

TESS GI Program Cycle I 2017

PRESENTATIONS

Contributed Talks

Magnetic Inflation and Stellar Mass of M dwarf stars
- Cool Stars 20, Boston, MA (2018)
- Know Thy Star - Know Thy Planet, Pasadena, CA (2017)

Is Lodén 1 an old and nearby open cluster?
- Tuesday Lunch Talk, Astronomy Department, The Pennsylvania State University, State College, PA (2013)

Poster Presentations

Simulation Tools for the Roman Space Telescope
- 245th American Astronomical Society Meeting, National Harbor, MD (2025)

Properties of the Roman WFI optical elements and comparisons with JWST/NIRcam, HST/WFC3, and 2MASS
- Accurate Flux Calibration workshop, Baltimore, MD (2024)

The Mass-Radius-Luminosity-Rotation Relationship for M Dwarf Stars
- 227th American Astronomical Society Meeting, Kissimmee, FL (2016)

Is Lodén 1 an old and nearby open cluster?
- 223rd American Astronomical Society Meeting, National Harbor, MD (2014)

Updates to the Exoplanet Orbit Database and Transit & Secondary Eclipse Ephemerides
- 221st American Astronomical Society Meeting, Long Beach, CA (2013)

The Exoplanet Orbit Database
- Undergraduate Planetary Science Research Conference, Lunar and Planetary Science Conference, The Woodlands, TX (2012)
- The 2012 Undergraduate Exhibition, The Pennsylvania State University, PA (2012)
- The 2011 Undergraduate Exhibition, The Pennsylvania State University, PA (2011)

TEACHING EXPERIENCE

Teaching Fellow at Boston University

- AS 105: Alien Worlds — Fall 2014
- AS 100: Cosmic Controversies — Spring 2015

Mentored Students

- Erica Sawczynec — Fall 2021 - Summer 2023
  - Graduate student at the University of Texas at Austin
  - Co-advising on the investigation of the H_2 emission in Class II Young Stellar Objects using IGRINS spectra

- Maryam Hussaini — Fall 2020
  - Former undergraduate student at the University of Texas at Austin, now a graduate student at Harvard University
  - Co-advised on the investigation on the impact of rotation velocity on measuring magnetic fields of K and M type stars. Resulted a publication of a RNAAS article.

- Brian Healy — Spring 2016 - Summer 2019
  - Former undergraduate student at Boston University, now a graduate student at Johns Hopkins University
  - Advised on the modeling the light curves and radial velocity measurements of eclipsing binaries. Resulted a publication of an AJ article.

SELECTED MEDIA APPEARANCES

- Can We Ever Understand the Size of Red Dwarf Stars?
  https://www.space.com/red-dwarf-stars-size-mystery.html
- Lodén 1 debunked
  https://www.salt.ac.za/news/loden-1-debunked/

SERVICE & PUBLIC OUTREACH

- Session chair at the Sagan Summer Workshop — July 2021
- Graduate Program Review Committee — Fall 2017 - Summer 2018
  - Served as a member of the committee to undertake an end-to-end review of the PhD graduate program of the Department of Astronomy at Boston University
- Public open night at Boston University — 2014 - 2018
  - Volunteered to run the weekly open night of the roof top observatory of the Department of Astronomy at Boston University for the community in Boston area
- Graduate Student Organization at Boston University — 2017
  - Representative of Department of Astronomy
- The Art of Astrophysics at Boston University — February 2016
  - Volunteered at the Art of Astrophysics, a competition for the BU community to create astronomy themed artwork/multimedia
  - Helped setting up, monitoring, and cleaning up the gallery
- Science by the Pint — January 2016
  - Volunteered at the event that is to connect scientific researchers with the members of the general public in the Boston area
  - Answered questions on exoplanets and their host-stars from the general public

HONORS & SCHOLARSHIPS

Anacapa Scholar, Thacher School, Ojai, CA	2015
Schreyer Honors College, The Pennsylvania State University	2010 - 2012
Elsbach Physics Scholarship, The Pennsylvania State University	2012
Kadke Family Endowed Scholarship, The Pennsylvania State University	2012
The President’s Freshman Award, The Pennsylvania State University	2009
Dean’s List, The Pennsylvania State University	2009 - 2012

PROFESSIONAL AFFILIATIONS

American Astronomical Society — November 2013 - Present

DATA and ANALYTICS & COMPUTER SKILLS

DATA and ANALYTICS Big Data Queries and Interpretation, Machine Learning Algorithms, Predictive Modeling, Data Mining and Visualization Tools

Languages Python, IDL, C++, SQL